

DDM: Dense Dynamics World Action Modeling for Physics-Sensitive Robotic Manipulation

Anonymous submission

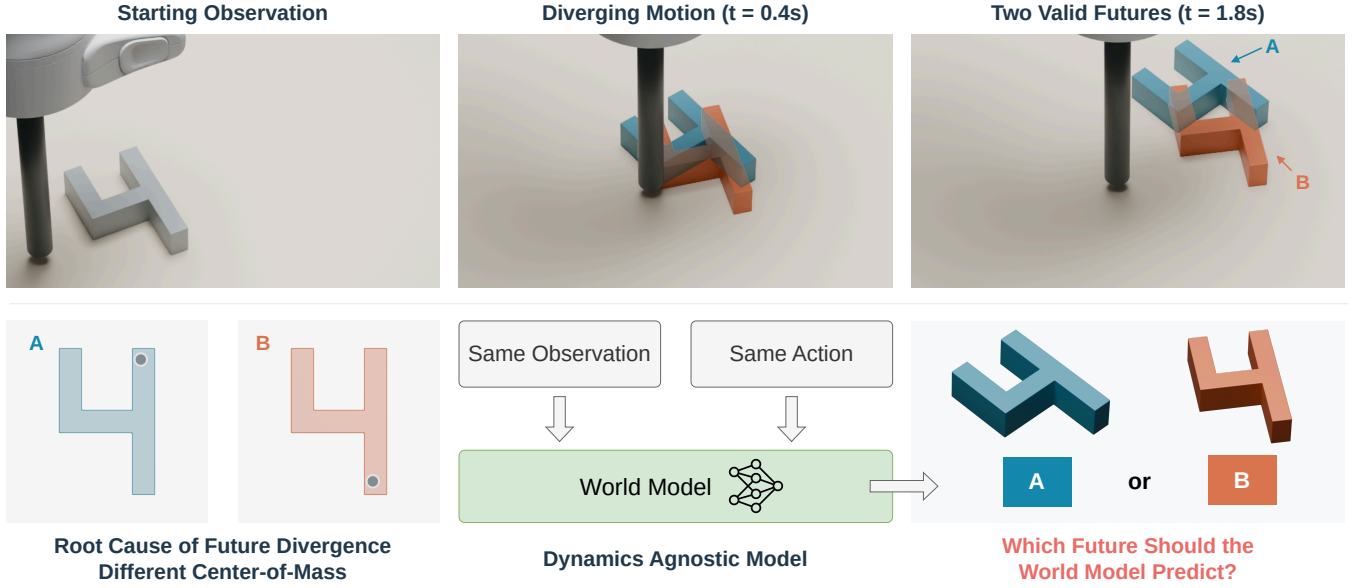


Fig. 1. Which future should a world model predict? Visually identical digits with different centers of mass follow different trajectories and reach different poses under the same push. A dynamics-blind model receives identical observations and actions in both cases. Colors and center-of-mass markers identify the illustrated dynamics; the arm visualizes the simulated pusher.

Abstract—Video generative models serve robotics as action-conditioned world models and goal-conditioned world-action models. Although their interfaces differ, both usually assume one fixed dynamics regime, i.e., an invariant action–outcome mapping. When dynamics change without altering appearance, identical actions yield different futures and identical goals require different actions. Conditioning on past interactions is therefore a natural response. Yet raw history provides only model-agnostic transition evidence: it records which outcome followed an action, but not how that outcome differs from what the model expected. Conventional action-only policies do not expose this distinction; video generative models do, through an explicit future prediction. We therefore introduce DDM, a dynamics-aware video-diffusion framework that converts this model-relative prediction error into context. DDM-cond compares realized futures with a frozen dynamics-blind reference and aggregates persistent latent offsets into a residual dynamics descriptor, without dynamics labels or a predefined parameterization. DDM-pred complements this context by packing transient observations into the existing latent budget. A shared architecture family supports separately trained WM and WAM modes. We evaluate both modes in simulation on pushing with latent center-of-mass shifts and tossing with latent actuation variations. Across both domains, interaction conditioning improves dynamics-sensitive prediction and adapts generated actions. On pushing at $k = 8$, residual conditioning improves dynamics identification from 31.96% to 44.25% over matched history, while DDM raises open-loop action success by 12.38 percentage points over matched Cosmos Policy. By coupling both mechanisms, DDM makes latent dynamics actionable for prediction and action generation.

I. INTRODUCTION

Video generative models increasingly support robotic manipulation in two complementary roles: action-conditioned world models (WM) predict how a scene will evolve, while goal-conditioned world-action models (WAM) jointly generate an action and the future it should realize [1]–[4]. Although their interfaces differ, both typically learn a single, fixed mapping between actions and outcomes. Dynamic manipulation breaks this assumption: latent changes in inertial or actuation dynamics can leave the scene visually unchanged while making the same action produce different futures and the same goal require different actions. Figure 1 exposes this ambiguity by replaying the same push from a bit-identical state under different dynamics. Across our counterfactual evaluation windows, final positions differ by a median of 7.97 cm and by as much as 19.61 cm. A dynamics-blind model must therefore reconcile incompatible transitions from identical visual evidence.

A common response is to condition policies or predictive models on past interactions [5]–[10]. Raw history, however, is model-agnostic: it records which outcome followed an action, but not how that outcome departs from the model’s own expectation. The learner must recover the persistent dynamics signal while discounting appearance, action geometry, and nominal motion; more fundamentally, the history does not distinguish what the model already explains from

what contradicts its transition prior.

Recent video-based policies make this distinction observable. Unlike an action-only policy, a video model emits an explicit future prediction that externalizes its learned action–outcome relation. Once the action is executed, the latent offset from prediction to realization reveals where, and by how much, that relation failed. Aggregating the persistent structure of this model-relative error across interactions can therefore identify how the active regime departs from the model’s prior. Video generation creates a second opportunity: the future target determines which dynamics evidence reaches the model. A terminal state may erase revealing motion after an object settles, whereas a dense trajectory preserves it.

We introduce DDM, a dynamics-aware video-diffusion framework built around these two observations. DDM-cond encodes latent offsets between realized futures and those predicted by a frozen dynamics-blind reference, then aggregates them into a residual dynamics descriptor (RDD) that modulates generation without dynamics labels. DDM-pred complements this model-relative context by packing observations across the prediction horizon into the existing future latent slot. We instantiate both components in separately trained WM and WAM instances of the same architecture—an interface we call *world action modeling*. Our contributions are:

- We pioneer video-based dynamics-aware world action modeling, unifying dynamics-aware future prediction and action generation within a common generative framework.
- We introduce DDM-cond, a model-relative conditioning mechanism that distills persistent dynamics mismatch from prior interactions.
- We introduce DDM-pred, a dense future objective that preserves transient dynamics evidence within the existing latent budget.
- We validate DDM on challenging push-digit and toss-cube benchmarks constructed from matched counterfactuals. Results show gains in dynamics-sensitive prediction, few-shot adaptation, and action generation across latent inertial and actuation shifts.
- We further show that DDM-pred improves the aggregate success of Cosmos Policy on 24 standard RoboCasa tasks, extending its benefit beyond dynamics-sensitive benchmarks.

II. RELATED WORK

Video models for robotic prediction and action generation. Action-conditioned video models predict future observations and enable planning through imagined rollouts [3], [4]. World-action models extend this interface by jointly generating actions and their anticipated outcomes [2]. This progression turns future prediction into an explicit account of the action–outcome relation encoded by the model. DDM uses that account as a reference for dynamics adaptation, while retaining separately trained WM and WAM modes within one architecture [1].

Dynamics conditioning and interaction-based adaptation. Dynamics-aware models have conditioned prediction or control on simulator parameters, spatial property maps, or learned dynamics latents [11]–[13]. Moving beyond explicit parameterization, dynamic manipulation systems infer hidden dynamics from prior transitions [5], [6], [14], [15]. IIDA aggregates raw transitions for implicit identification [7], whereas IPA Policy conditions one-shot action prediction on a partial probing interaction [9]. DyWA further adapts joint action–future prediction from history, while supervising its embedding with privileged geometry and dynamics features [8]. These methods establish interaction history as a powerful adaptation signal, but represent it either through privileged supervision or as absolute transition evidence. Direct history can also induce copycat or causal-confusion effects [16], [17]. Our IIDA+ and IPA Policy+ baselines scale the corresponding history-conditioning principles to pretrained WM and WAM backbones with k complete interactions. DDM receives the same observations, actions, and realized futures, but expresses each transition relative to a frozen reference model’s expected future. It therefore isolates the mismatch relevant to adaptation without estimating an explicit dynamics parameter.

Prediction residuals as adaptation signals. Prediction error is a classical identification signal [18] and has more recently supported ballistic correction, world-model adaptation, and residual dynamics learning [19]–[22]. These approaches use residuals primarily as correction signals for nominal predictions or learned transitions. DDM instead treats prediction error as a conditioning representation: it aggregates latent discrepancies from completed interactions before generating subsequent futures and actions. Residual structure that persists across interactions from one regime captures how its dynamics depart from the frozen reference, while aggregation attenuates transition-specific variation.

III. METHOD: DDM

A. Problem formulation

Let o_t be the visual observation at query time t , and let a_t denote the action representation executed from that state. In our controlled domains, a_t compactly parameterizes a manipulation primitive rather than a temporal action sequence. Let H denote the future-prediction horizon, and let $o_{t,H}$ be the corresponding future observation target assembled from frames in $(t, t + H]$. The frozen video tokenizer $\mathcal{E}$ maps the current observation and future target to $z_t = \mathcal{E}(o_t)$ and $z_{t,H} = \mathcal{E}(o_{t,H})$, respectively. When t and H are clear, we write $z^+ \equiv z_{t,H}$. Let ξ index the unobserved dynamics regime and let p_ξ denote its controlled transition distribution. The ambiguity motivating DDM is that different regimes induce different transitions even when the observation and action are identical:

$$z^+ \sim p_\xi(z^+ | z_t, a_t). \quad (1)$$

The robot neither observes ξ nor receives its identity as a label. Instead, the query is accompanied by an ordered history of k completed interactions from the same persistent

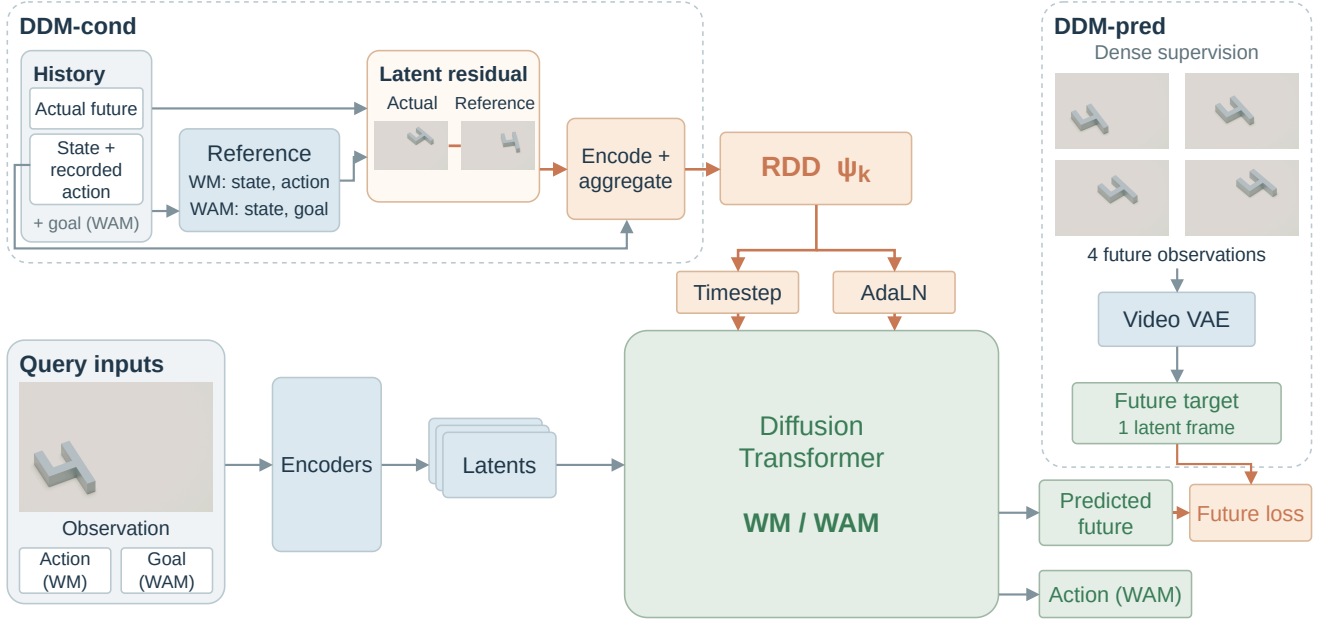


Fig. 2. **DDM combines residual conditioning and dense future supervision.** Completed interactions are compared with frozen WM/WAM references; their future residuals are encoded and aggregated into the RDD ψ_k^j , which modulates generation. DDM-pred packs four observations across the prediction horizon into the existing future latent component. Only the future loss is shown; rendered states are schematic.

regime. The i -th interaction contains its starting latent z_i , recorded action a_i , and realized future latent z_i^+ . In WAM mode, g_i is the known visual goal that elicited a_i . DDM summarizes this history as a mode-specific residual dynamics descriptor (RDD) ψ_k^j , where $j \in \{\text{WM}, \text{WAM}\}$. WM uses the tuples (z_i, a_i, z_i^+) ; WAM additionally uses g_i to query its frozen joint reference. Rather than estimating ξ , the descriptor need only retain persistent model-relative mismatch useful for adapting prediction or action generation. We omit j when the mode is unambiguous; Sec. III-C defines the encoder and aggregation explicitly.

B. Architecture and modeling modes

Figure 2 gives an overview of the two dynamics-aware components. Each mode instantiates a 2B-parameter Cosmos-Predict2 video diffusion transformer. Following the latent token injection design of Cosmos Policy [2], the normalized action representation is tiled over one video-latent frame. Its spatiotemporal VAE compresses four raw frames into one latent frame. The resulting sequence contains four components—a fixed-zero blank placeholder, current observation, action, and future target—with mode-specific masks; the placeholder is excluded from the loss.

Denosing follows the EDM formulation [23]. In Eq. (2), $\mathbf{x}_0$ is the clean sequence; m and ℓ are the conditioning and loss masks; $\mathbf{c}$ is the language embedding; and $\mathbf{n}$ is isotropic Gaussian noise at level σ :

$$\tilde{\mathbf{x}} = m \odot \mathbf{x}_0 + (1 - m) \odot (\mathbf{x}_0 + \mathbf{n}),$$

$$\mathcal{L}_{\text{DDM}} = \mathbb{E} \left[w(\sigma) \|\ell \odot (D_\theta(\tilde{\mathbf{x}}; \sigma, \mathbf{c}, \psi_k) - \mathbf{x}_0)\|_2^2 \right]. \quad (2)$$

where $w(\sigma)$ is the EDM weight and D_θ is the preconditioned denoiser. The masks separately specify observed and supervised components; σ enters through adaptive layer normalization, $\mathbf{c}$ through cross-attention, and ψ_k through modulation. The expectation covers data and context draws, diffusion noise, and descriptor dropout.

The two modes capture how dynamics affect both the outcome of a given action and the action required for a desired outcome. Their factorizations are

$$\text{world model: } p_{\theta_{\text{WM}}}(z^+ | z_t, a_t, \psi_k), \quad (3)$$

$$\text{world-action model: } p_{\theta_{\text{WAM}}}(a_t, z^+ | z_t, g, \psi_k), \quad (4)$$

where g is the visual goal. In the WAM latent layout, (z_t, g) denotes the current observation rendered with the goal marker and encoded into one frame, rather than two latent slots. Mode-specific masks implement the two factorizations: WM conditions on the current observation and action and supervises the future, whereas WAM conditions on the current observation and goal and supervises the jointly generated action and future. We use *world action modeling* for this broader generative interface, rather than for a single jointly shared model: θ_{WM} and θ_{WAM} are optimized separately and use distinct frozen references and residual stores.

C. DDM-cond: Residual dynamics conditioning

A residual is meaningful only relative to a stable expectation. Each mode therefore uses a frozen dynamics-blind reference—Cosmos-Predict2 for WM and Cosmos Policy for WAM—from which we warm-start the corresponding DDM model. Training then jointly optimizes the encoder, modulation layers, and backbone. Both references are trained

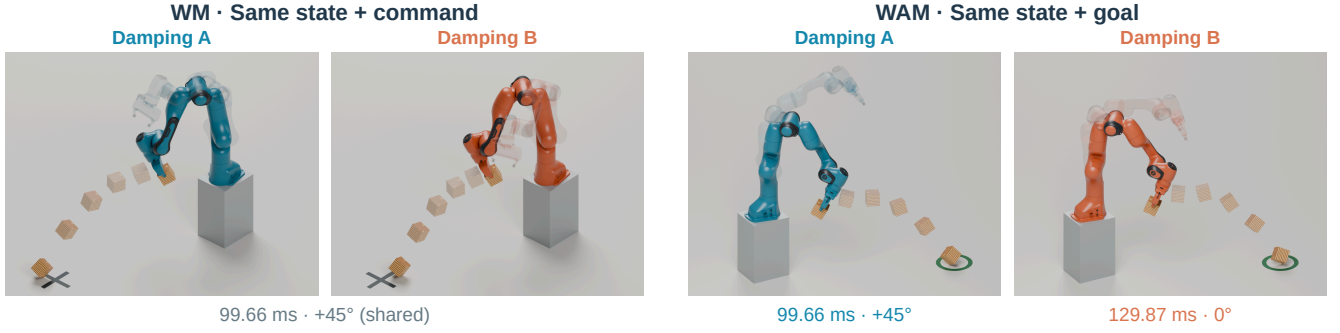


Fig. 3. **Cube tossing under latent actuation dynamics.** *Left pair (WM):* the same initial state and command produce different motion under two damping settings. *Right pair (WAM):* the same state and goal require different release-time/yaw commands. Crosses provide a fixed reference; rings mark the goal. Solid arms show release and faded arms show follow-through; colors identify damping. Grasp, release, and cube flight are schematized from recorded motion, ending at first contact.

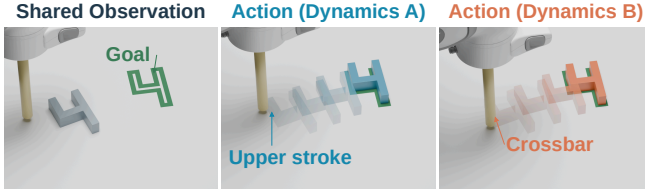


Fig. 4. **Goal-conditioned digit pushing.** Different hidden mass distributions require different push primitives to reach one target from the same observation. Faded and solid digits show motion and final pose; green outlines mark the goal.

without interaction context and therefore marginalize over the training dynamics.

Let $f_{\bar{\phi}_{\text{WM}}}$ and $f_{\bar{\phi}_{\text{WAM}}}$ denote the distinct frozen references, and let $\mathcal{S}_{\text{WM}}$ and $\mathcal{S}_{\text{WAM}}$ be their fixed sampling operators. For interaction i ,

$$\hat{z}_i^{\text{ref}, \text{WM}} = \mathcal{S}_{\text{WM}}(f_{\bar{\phi}_{\text{WM}}}; z_i, a_i, \epsilon_0), \quad (5)$$

$$(\hat{a}_i^{\text{ref}, \text{WAM}}, \hat{z}_i^{\text{ref}, \text{WAM}}) = \mathcal{S}_{\text{WAM}}(f_{\bar{\phi}_{\text{WAM}}}; z_i, g_i, \epsilon_0), \quad (6)$$

$$\delta_i^j = z_i^+ - \hat{z}_i^{\text{ref}, j}. \quad (7)$$

Here $j \in \{\text{WM}, \text{WAM}\}$ and ϵ_0 is the fixed sampler-noise realization for the stored interaction; the checkpoint, sampler, and noise seed are shared across matched arms, defining a stable model-relative coordinate.

For WM, the observed and predicted futures share a_i , so δ_i^{WM} is a matched-action transition discrepancy. WAM introduces a second source through its blind joint reference, $p_{\bar{\phi}_{\text{WAM}}}(a, z^+ | z, g) = p(a | z, g)p(z^+ | z, g, a)$. Let the realized draw be $z_i^+ = F_{\xi}(z_i, a_i)$, and let $\bar{F}(z, g, a)$ denote a fixed-sampler conditional future coordinate under the second factor. Because the observed future follows a_i but the reference future accompanies $\hat{a}_i^{\text{ref}, \text{WAM}}$, defining $\nu_i = \bar{F}(z_i, g_i, \hat{a}_i^{\text{ref}, \text{WAM}}) - \hat{z}_i^{\text{ref}, \text{WAM}}$ yields the accounting identity

$$\begin{aligned} \delta_i^{\text{WAM}} &= \Delta_i^{\text{trans}} + \Delta_i^{\text{act}} + \nu_i, \\ \Delta_i^{\text{trans}} &= F_{\xi}(z_i, a_i) - \bar{F}(z_i, g_i, a_i), \\ \Delta_i^{\text{act}} &= \bar{F}(z_i, g_i, a_i) - \bar{F}(z_i, g_i, \hat{a}_i^{\text{ref}, \text{WAM}}). \end{aligned} \quad (8)$$

Here Δ_i^{trans} is matched-action transition mismatch, Δ_i^{act} reflects the reference's different action choice, and ν_i retains joint-sampling variation. DDM does not identify them separately: an individual δ_i^{WAM} is model-relative interaction evidence, not a pure estimate of ξ . Its aggregate ψ_k^{WAM} becomes dynamics context by retaining mismatch that persists across shared-regime interactions.

A mode-specific encoder E_j and projection P_j aggregate these residuals:

$$h_i^j = E_j(z_i, a_i, \delta_i^j), \quad \psi_k^j = P_j([b_1 h_1^j; \dots; b_k h_k^j]), \quad (9)$$

where h_i^j is the per-interaction embedding and $b_i \in \{0, 1\}$ masks absent entries. Each interaction field is projected to 256 dimensions before a 512-wide residual trunk; the k ordered, masked slots are concatenated and projected to the 256-dimensional descriptor. Selected interactions retain each task sampler's deterministic order, shared across matched arms; any padding occupies trailing slots and is zero-masked. The descriptor is learned end-to-end through the denoising objective in Eq. (2); no auxiliary dynamics-label loss is used. The matched *two-stage history* control replaces δ_i^j with z_i^+ in Eq. (9); it is distinct from the single-stage IIDA+ and IPA Policy+ baselines.

The resulting ψ_k modulates generation through two pathways in each transformer block rather than through the latent sequence. Let s index latent frames, $e_{\sigma, s}$ denote the noise-level embedding, and r_s the adaptive-layer-normalization/low-rank-adaptation (AdaLN-LoRA) modulation vector. Learned MLPs M_e and M_r project the descriptor, while α_s combines the task-specific fixed scale and frame mask specified in Sec. IV-C:

$$\begin{aligned} e'_{\sigma, s} &= e_{\sigma, s} + \alpha_s M_e(\psi_k), \\ r'_s &= r_s + \alpha_s M_r(\psi_k). \end{aligned} \quad (10)$$

The primed quantities replace their unmodulated counterparts in every block, forming a FiLM-like route through AdaLN [24], [25]. The roughly 100M added parameters amount to 5% of the backbone.

Every context interaction precedes the query. The reference prediction is discarded after subtraction, and the WAM



Fig. 5. **Selected future-prediction examples.** Actual futures (left), predictions (center), and error maps (right). Rows show Cosmos-Predict2 (blind reference), then DDM, at 1.40 s (push) and 1.20 s (toss), with identical crops within each task. Darker maps indicate larger mean absolute RGB error (8-bit units; common scale saturated at 64).

TABLE I
PUSH-DIGIT WM MODE RESULTS.

	Dynamics identification (%) $\uparrow$		
	Training	All five	Held-out ($k = 8$)
Cosmos-Predict2	25.00	—	—
Two-stage hist., $k = 2$	28.00	21.67	—
Two-stage hist., $k = 4$	29.96	21.87	—
Two-stage hist., $k = 8$	31.96	23.83	16.83
DDM, $k = 2$	37.58	29.00	—
DDM, $k = 4$	41.46	32.20	—
DDM, $k = 8$	44.25	34.90	24.33

TABLE II
TOSS-CUBE WM MODE RESULTS.

	Warm start	WMSE $\downarrow$	Yaw identification (%) $\uparrow$
			(chance 14.29)
Cosmos-Predict2	pretrained	0.0437	14.29
IIDA+, $k = 8$	pretrained	0.0297	56.55
Two-stage hist., $k = 2$	reference	0.0268	63.69
Two-stage hist., $k = 4$	reference	0.0269	60.71
Two-stage hist., $k = 8$	reference	0.0269	62.50
DDM, $k = 2$	reference	0.0257	63.69
DDM, $k = 4$	reference	0.0255	63.69
DDM, $k = 8$	reference	0.0250	66.67

goal marker appears only in the conditioned observation, never in either future being subtracted.

D. DDM-pred: Dense future prediction

Whereas DDM-cond changes how past interactions are represented, DDM-pred determines which dynamics evidence reaches WAM through its training target. World models conventionally predict a temporally dense future, but world-action models often supervise only a terminal outcome. Cosmos Policy, for example, repeats the terminal observation to fill its future slot [2]. Within the same architecture and spatiotemporal encoder, DDM-pred changes only this target, replacing terminal repeats with observations distributed across the prediction horizon:

$$z_{t,H} = \mathcal{E}\left(\left(o_{t+\lfloor qH/4 \rfloor}\right)_{q=1}^4\right). \quad (11)$$

The resulting short trajectory preserves motion that may disappear after the object settles while occupying the existing future component of $\mathbf{x}_0$ in Eq. (2); it therefore changes neither the latent budget nor the action-generation interface. An architecture-matched RoboCasa check (Sec. IV-F) indicates that the denser target does not trade away aggregate policy success. We use DDM-pred to name this target change in WAM; WM already uses the same dense target by default. Unless stated, DDM uses DDM-cond in both modes, retains standard dense supervision for WM, and applies DDM-pred to WAM. Each WAM reference is a context-free Cosmos Policy instance matched to its arm’s future target: the default dense model and the repeated-terminal ablation therefore use

separately trained references and residual stores. Within each store, realized and reference futures use the same encoding.

IV. EXPERIMENTS

We construct matched counterfactuals that remove visual cues to dynamics. A WM must distinguish multiple valid futures for an identical state–action query, whereas a WAM must generate different actions for an identical state–goal query. The two domains use separate banks and metrics; comparisons are made only within each protocol.

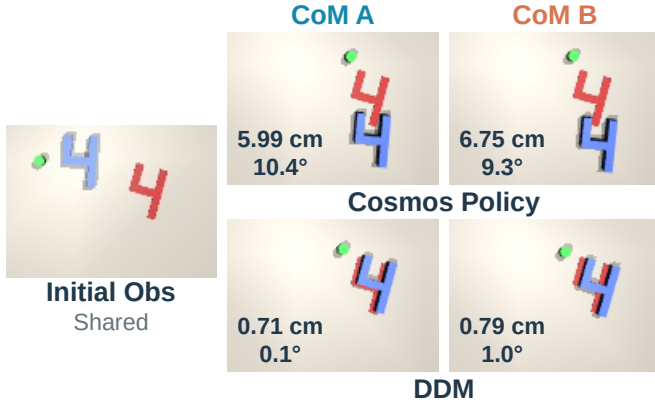
A. Tasks and counterfactual datasets

Task 1: planar pushing under latent inertial dynamics.

A cylindrical pusher strikes a seven-segment digit whose invisible ballast carries 90% of its mass at one of five positions spaced 1.75 cm apart. The same state and push can therefore produce different translations and rotations, challenging WM prediction, while one goal requires different approach angles, speeds, and lateral offsets from the WAM (Fig. 4). Models train on four ballast positions. The WM protocol additionally evaluates $y+1.75$ as an unseen dynamics value; the WAM protocol evaluates held-out goal windows over the four training values.

Task 2: dynamic tossing under latent actuation dynamics. A Franka swings and releases an 8 cm cube; we vary swing- and yaw-joint damping over 35 visually indistinguishable combinations. Under a shared command, WM must predict damping-dependent flight; for a shared landing

(C) Push Digit World Action Model Task



(D) Toss Cube World Action Model Task

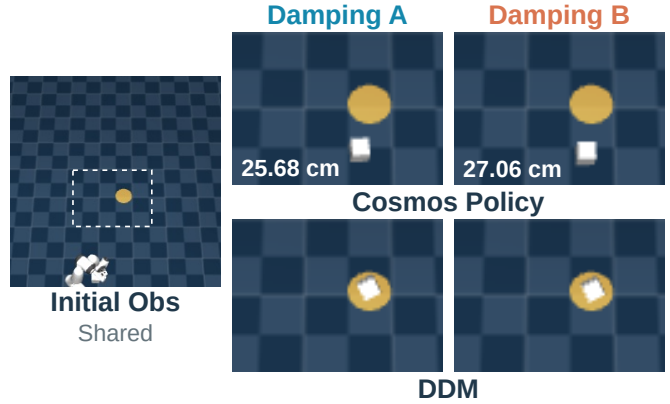


Fig. 6. **Selected goal-reaching examples.** Shared start and goal (left); Cosmos Policy/DDM outcomes in the top/bottom rows under dynamics A/B (center/right). Push: terminal observations after one open-loop action, with position/yaw error (cm/degrees); red target, green pusher. Toss: first-contact replays of snapped actions, with actual distance to the yellow goal center (cm), not regret. Dashed box: matched outcome crop.

TABLE III
PUSH-DIGIT WAM MODE RESULTS.

	Success (%) $\uparrow$	95% CI	Δ vs CP (pp; p)
DDM, $k = 2$	55.13	[50.8,59.5]	+9.9 (2e-07)
DDM, $k = 4$	55.75	[51.3,60.1]	+10.5 (1e-07)
DDM, $k = 8$	57.63	[53.4,61.9]	+12.4 (2.6e-10)
DDM $k = 8$, RDD=0	45.25	—	—
Two-stage hist., $k = 2$	46.37	[41.6,51.0]	+1.1 (0.57)
Two-stage hist., $k = 4$	45.75	[41.4,50.1]	+0.5 (0.84)
Two-stage hist., $k = 8$	46.75	[42.4,51.1]	+1.5 (0.46)
IPA Policy+, $k = 8$	54.13	—	—
Dynamics oracle	75.12	[71.1,79.0]	+29.9 (4e-47)
Cosmos Policy (matched)	45.25	[40.8,49.8]	—

TABLE IV
TOSS-CUBE WAM MODE RESULTS.

	Unseen goals (cm; $n = 480$)	Unseen dynamics (cm; $n = 616$)
Cosmos Policy (matched)	10.42	7.08
Two-stage hist., $k = 2$	1.29	3.03
Two-stage hist., $k = 4$	1.29	3.05
Two-stage hist., $k = 8$	1.36	3.05
DDM, $k = 2$	1.40	3.08
DDM, $k = 4$	1.36	3.03
DDM, $k = 8$	1.38	2.95

goal, WAM must adapt release time and yaw (Fig. 3). The WM split withholds one swing-damping row and identifies the seven yaw-damping values. The WAM split evaluates unseen goals over 24 trained regimes and unseen dynamics over 11 held-out regimes.

RoboCasa dense-target check. Outside controlled dynamics, we isolate DDM-pred on 24 RoboCasa tasks, including pick-and-place [26]. The comparison changes only Cosmos Policy’s target to Eq. (11); both arms train for 100,000 iterations on eight H100 GPUs. We report per-task and macro-average success from this descriptive, single-checkpoint comparison.

Controlled data come from MuJoCo counterfactual banks [27], which roll each serialized transition through every dynamics value. The push WM bank contains 200 shards of 120 transitions, split 160/40 by shard. We provide $k \in \{2, 4, 8\}$ same-regime interactions; grouping occurs only after the split, excludes the query, and supplies no identity or dynamics label to the model.

B. Metrics and baselines

Dynamics identification. Let $\mathcal{V}$ be the tested regimes, $\hat{z}_v$ the prediction conditioned on regime v , and z_w^* the counterfactual future under w . With $d_{v,w} = \|\hat{z}_v - z_w^*\|$, we report $\Pr[\arg \min_{w \in \mathcal{V}} d_{v,w} = v]$; chance is $1/|\mathcal{V}|$. Matched variants share batch position and sampler noise, so blind

predictions score exactly chance. This tests sensitivity, not recovery of a named parameter. For push, $\mathcal{V}$ contains the four training regimes or all five regimes; toss identifies seven yaw-damping values within the held-out swing-damping row. **Future-prediction error.** Push uses per-element latent mean-squared error (MSE). Toss reports windowed latent MSE (WMSE) within a 14×14 region centered identically for all arms on the true cube, preventing the static background from dominating the score.

Push-digit action success. One sampled parameterized action is executed open-loop and succeeds within 1.5 cm and 2° of the goal; expert replay succeeds on 99.75% of rows. Success rates are percentages and differences use percentage points (pp).

Toss-cube landing regret. Generated actions are snapped to the nearest of 180 pre-rolled actions and scored by oracle-relative landing regret in the corresponding dynamics regime; lower is better. The lookup protocol has an empirical floor of 1.2 cm.

Cosmos-Predict2 and Cosmos Policy (CP). These dynamics-blind WM and WAM references, respectively, receive no interaction context.

IIDA+. This scaled IIDA baseline jointly trains a history encoder with the pretrained WM using k complete interactions.

IPA Policy+. This scaled IPA Policy baseline replaces the original partial probe [9] with k complete interactions. For

both “+” baselines, the suffix denotes our larger pretrained adaptation, not an exact reproduction.

Two-stage history. This matched RDD control uses the same reference warm start, store, encoder path, context budget, and optimization as DDM, but replaces each residual with the realized future. We compare it with DDM for every $k \in \{2, 4, 8\}$.

Dynamics oracle. Ground-truth dynamics enter through the same modulation interface, exposing the remaining gap to perfect dynamics context without changing the WAM consumer.

C. Implementation and statistical analysis

All modes use 224×224 inputs. Inference uses 20 denoising steps for push WM and 35 otherwise, with guidance 1.5. Push uses $\alpha_s = 1$ on all frames; toss modulates generated frames only, at 0.1 for WM and 0.029 for WAM. Samplers and references are fixed within comparisons, and each residual store caches one offline reference sample per interaction.

Push WM evaluates 600 matched windows across four training regimes (2,400 trials) and, when included, all five regimes (3,000); push WAM uses 800 rows from 200 goal windows. Toss WM uses 168 rows; toss WAM uses 480 unseen-goal and 616 unseen-dynamics pairs. Intervals and paired differences cluster by matched window, goal, or action. Push-action p -values use paired McNemar tests; all 12 rejected nulls remain rejected after Holm correction over the 20 planned tests.

Matched push WAM models are read at iteration 10,000; IPA Policy+ uses 12,000 and is descriptive. For toss WM (Table II), Cosmos-Predict2, IIDA+, and reference-warm-started rows use iterations 3,000, 7,500, and 10,000. Each model is trained once, so intervals quantify evaluation-unit, not training-run, variation; k , regime, mode, and sampler-seed comparisons probe separate robustness axes.

D. Model-relative residuals identify the active dynamics

On push-digit WM, prediction residuals provide more useful dynamics context than the same transitions encoded directly. DDM improves identification at every context budget (Table I). At $k = 8$, accuracy rises from 31.96% (95% confidence interval (CI) [30.71, 33.08]%) with matched history to 44.25% ([42.75, 45.83]%), versus 25.00% chance; the paired gain is +12.29 percentage points ([+10.33, +14.21]). The matched- k advantage grows from +9.58 pp at $k = 2$ to +12.29 pp at $k = 8$. At $k = 2, 4$, DDM also lowers per-element latent MSE from 0.0184, 0.0181 to 0.0158, 0.0133. Figure 5 shows selected large localization improvements for push ballast $y = -1.75$ cm and toss ($b_{\text{swing}}, b_{\text{yaw}}$) = (13, 10). Pixel error is distinct from the latent RDD; examples are illustrative.

E. Residual context extends beyond training dynamics

The RDD is not tied to a catalog of training labels. At $k = 8$, adding the held-out $y + 1.75$ regime raises DDM identification to 34.90%, versus 23.83% for history and 20.00% chance; on that slice alone, DDM scores 24.33%

versus 16.83%. The value is excluded from training, while the query is conditioned on interactions from the same held-out regime. This experiment therefore measures few-shot adaptation and shows that residual context remains informative beyond the training dynamics.

F. Dense targets preserve transient dynamics evidence

Dense future targets expose dynamics evidence along the trajectory rather than only after the object settles. For toss-cube WM with DDM at $k = 4$, yaw-damping identification is 63.69% from the four-frame future but only 17.26% from its terminal frame (Cosmos-Predict2: 14.29%), localizing the useful evidence to the motion. The direct target comparison is more conservative: at the shared $t = 1.2$ s frame, dense and repeated-terminal training are unresolved in raw pixel error, while correcting each target for its VAE reconstruction floor favors dense prediction on 101/168 rows. RoboCasa provides a separate target-design check without DDM-cond or controlled dynamics shifts. There, dense targets improve 18/24 tasks, tie one, and raise macro-average success from 37.9% to 40.7% (Table V), showing that the same supervision also benefits a regular generalist-policy setting.

G. Model-relative context changes the generated action

On push-digit, model-relative interaction context changes the generated action under an otherwise identical goal query. Across all $k \in \{2, 4, 8\}$, DDM exceeds matched history by 8.75–10.88 pp (Table III). At $k = 8$, DDM reaches 57.63%, or +12.38 pp over matched Cosmos Policy ($p = 2.6e - 10$). IPA Policy+ reaches 54.13% descriptively, as its recipe and read point differ. Evaluating the same DDM $k = 8$ checkpoint with its RDD zeroed yields 45.25%, a 12.38 pp reduction ($p = 3.1e - 11$); its equality with the matched Cosmos Policy rate is numerical rather than definitional. The DDM–history gain remains +10.50 pp at extreme dynamics and averages +10.04 pp across three sampler seeds, which do not measure training variability. A targeted repeated-terminal ablation uses only the endpoint budgets $k \in \{2, 8\}$ and 800 held-out rows; intervals are clustered over 200 goal windows. DDM still beats matched history by +12.50 pp at $k = 8$ ([+9.00, +16.00]) and +5.00 pp at $k = 2$ ([+1.38, +8.50]). Dense and terminal scores are read against different residual-reference checkpoints and expert-replay ceilings, so we report them as parallel diagnostics rather than a single causal comparison. Figure 6 visualizes representative push and toss outcomes generated from shared starts and goals.

H. Interaction conditioning transfers to actuation dynamics

Toss-cube asks whether the same interface transfers from an object’s latent inertia to the robot’s latent actuation dynamics. In WAM mode (Table IV), interaction conditioning cuts unseen-goal landing regret from 10.42 cm to 1.29–1.40 cm and unseen-dynamics regret from 7.08 cm to 2.95–3.08 cm, winning 386–398 of 480 unseen-goal pairs. For the two evaluated history checkpoints, zeroing removes 71–75% of the gain. Both representation families are insensitive to k ;

TABLE V
ROBOCASA WAM MODE RESULTS.

Task	CP (%)	DDM-pred (%)	Δ (pp)	Task	CP (%)	DDM-pred (%)	Δ (pp)	Task	CP (%)	DDM-pred (%)	Δ (pp)
PnP Counter→Cab.	22.0	17.3	-4.7	Open single door	42.7	48.0	+5.3	Turn on sink faucet	70.7	46.7	-24.0
PnP Cab.→Counter	9.3	12.7	+3.4	Close single door	64.7	81.3	+16.6	Turn off sink faucet	72.7	79.3	+6.6
PnP Counter→Sink	16.0	23.3	+7.3	Open double door	48.7	71.3	+22.6	Turn sink spout	78.0	78.7	+0.7
PnP Sink→Counter	25.3	27.3	+2.0	Close double door	45.3	52.0	+6.7	Coffee: setup mug	8.7	8.7	0.0
PnP Counter→Micro.	4.7	10.7	+6.0	Open drawer	44.0	51.3	+7.3	Coffee: serve mug	32.7	34.0	+1.3
PnP Micro.→Counter	13.3	19.3	+6.0	Close drawer	93.3	95.3	+2.0	Coffee: press button	40.0	54.7	+14.7
PnP Counter→Stove	6.7	11.3	+4.6	Turn on stove	34.0	36.0	+2.0	Turn on microwave	33.3	32.0	-1.3
PnP Stove→Counter	9.3	12.0	+2.7	Turn off stove	17.3	16.7	-0.6	Turn off microwave	77.3	58.0	-19.3
Mean (24 tasks)	37.9	40.7	+2.8	Improved / tied / lower: 18 / 1 / 5							

on unseen goals they plateau near the 1.2 cm candidate-bank floor, and their pairwise differences never exceed 0.11 cm. Toss-cube therefore establishes transfer of the interaction-conditioning interface, while push-digit resolves the representation comparison.

The WM results show the same transfer. Across $k \in \{2, 4, 8\}$, toss-cube DDM lowers windowed latent MSE without reducing identification (Table II). At $k = 8$, it obtains 0.0250 MSE and 66.67% identification, versus 0.0269 and 62.50% for history; an action-clustered wild bootstrap over 24 actions gives $p = 0.0033$ for the +0.0019 MSE gap.

The advantage is concentrated where prior actions expose the dynamics shift: within the context action range, its MSE advantage grows to +0.0036, with wins on 68 of 84 rows.

V. CONCLUSION

DDM addresses manipulation under latent dynamics from two complementary directions. DDM-cond derives model-relative interaction context from what a frozen dynamics-blind reference repeatedly fails to explain, while DDM-pred retains the transient future evidence through which such mismatch is learned. Push-digit shows that residual context improves dynamics identification over matched history and raises open-loop success over Cosmos Policy. Toss-cube transfers the same interaction-conditioning interface from latent inertia to actuation dynamics, while a separate RoboCasa check shows the target change leaves aggregate policy success intact. The resulting world action modeling framework makes both future prediction and action generation responsive to the active dynamics.

REFERENCES

- [1] Z. Wang, K. Sivakumar, J. Shang, Y. Hu, Z. Xie, R. Gong, X. Zhang, and K. Schmeckpeper, "Efficient sim-to-real transfer of world-action models from synthetic priors," in *CVPR Embodied AI Workshop*, 2026, arXiv:2606.31101.
- [2] M. J. Kim, Y. Gao, T.-Y. Lin, Y.-C. Lin, Y. Ge, G. Lam, P. Liang, S. Song, M.-Y. Liu, C. Finn, and J. Gu, "Cosmos policy: Fine-tuning video models for visuomotor control and planning," *arXiv preprint arXiv:2601.16163*, 2026.
- [3] S. Li, Y. Gao, D. Sadigh, and S. Song, "Unified video action model," in *Robotics: Science and Systems (RSS)*, 2025, arXiv:2503.00200.
- [4] G. Zhou, H. Pan, Y. LeCun, and L. Pinto, "DINO-WM: World models on pre-trained visual features enable zero-shot planning," *arXiv preprint arXiv:2411.04983*, 2024.
- [5] A. Kumar, Z. Fu, D. Pathak, and J. Malik, "RMA: Rapid motor adaptation for legged robots," in *Robotics: Science and Systems XVII*, 2021, arXiv:2107.04034.
- [6] H. Qi, A. Kumar, R. Calandra, Y. Ma, and J. Malik, "In-hand object rotation via rapid motor adaptation," in *Proc. 6th Conf. on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 205, 2022, pp. 1722–1732.
- [7] B. Evans, A. Thankaraj, and L. Pinto, "Context is everything: Implicit identification for dynamics adaptation," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, Philadelphia, PA, USA, 2022, pp. 2642–2648.
- [8] J. Lyu, Z. Li, X. Shi, C. Xu, Y. Wang, and H. Wang, "DyWA: Dynamics-adaptive world action model for generalizable non-prehensile manipulation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 11 058–11 068.
- [9] Z. Wang and A. H. Qureshi, "Implicit physics-aware policy for dynamic manipulation of rigid objects via soft body tools," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, arXiv:2502.05696.
- [10] X. Zhang, S. Liu, P. Huang, W. J. Han, Y. Lyu, M. Xu, and D. Zhao, "Dynamics as prompts: In-context learning for sim-to-real system identifications," *arXiv preprint arXiv:2410.20357*, 2024.
- [11] Y. Li, H. Wen, B. Song, J. Zhu, A. Hao, C. Chen, J. Chen, J. Wu, X. Chu, and M. Wang, "Learning explicit physical parameter control and benchmarking for video generation," *arXiv preprint arXiv:2607.18924*, 2026.
- [12] S. Narayanan, Z. Jiang, S. Narasimhan, and M. Chandraker, "PhyCo: Learning controllable physical priors for generative motion," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, arXiv:2604.28169.
- [13] Y. Shen, J. Xiong, T. Yu, and I. Lourntzou, "Phantom: Physics-infused video generation via joint modeling of visual and latent physical dynamics," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, arXiv:2604.08503.
- [14] J. Li, W. S. Lee, and D. Hsu, "Push-net: Deep planar pushing for objects with unknown physical properties," in *Proceedings of Robotics: Science and Systems (RSS)*, Pittsburgh, Pennsylvania, USA, Jun. 2018.
- [15] M. Bauza, F. Alet, Y.-C. Lin, T. Lozano-Pérez, L. P. Kaelbling, P. Isola, and A. Rodriguez, "Omnipush: Accurate, diverse, real-world dataset of pushing dynamics with RGB-D video," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, Macau, China, 2019, pp. 4265–4272.
- [16] C. Wen, J. Lin, T. Darrell, D. Jayaraman, and Y. Gao, "Fighting copycat agents in behavioral cloning from observation histories," in *Advances in Neural Information Processing Systems*, vol. 33. Curran Associates, Inc., 2020, pp. 2564–2575, arXiv:2010.14876.
- [17] P. de Haan, D. Jayaraman, and S. Levine, "Causal confusion in imitation learning," in *Advances in Neural Information Processing Systems*, vol. 32. Curran Associates, Inc., 2019, arXiv:1905.11979.
- [18] R. Isermann, "Process fault detection based on modeling and estimation methods — a survey," *Automatica*, vol. 20, no. 4, pp. 387–404, 1984.
- [19] A. Zeng, S. Song, J. Lee, A. Rodriguez, and T. Funkhouser, "Tossing-Bot: Learning to throw arbitrary objects with residual physics," *IEEE Transactions on Robotics*, vol. 36, no. 4, pp. 1307–1319, Aug. 2020, arXiv:1903.11239.
- [20] T. An, J. Jia, G. Li, J. Li, C. Zhou, P. Liu, B. Lyu, J. Bai, X. Guo, G. Li, and J. Yang, "Feedback world model enables precise guidance of diffusion policy," *arXiv preprint arXiv:2605.15705*, 2026.
- [21] S. Patel, K. Zhang, S. Pokkali, S. Lazebnik, and Y. Li, "Learning

physics-guided residual dynamics for deformable object simulation,” *arXiv preprint arXiv:2607.13451*, 2026.

- [22] J. Lanier, K. Kim, A. Karamzade, Y. Liu, A. Sinha, K. He, D. Corsi, and R. Fox, “Adapting world models with latent-state dynamics residuals,” *arXiv preprint arXiv:2504.02252*, 2025.
- [23] T. Karras, M. Aittala, T. Aila, and S. Laine, “Elucidating the design space of diffusion-based generative models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022, arXiv:2206.00364.
- [24] W. Peebles and S. Xie, “Scalable diffusion models with transformers,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 4172–4182, arXiv:2212.09748.
- [25] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, “FiLM: Visual reasoning with a general conditioning layer,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018, arXiv:1709.07871.
- [26] S. Nasiriany, A. Maddukuri, L. Zhang, A. Parikh, A. Lo, A. Joshi, A. Mandlekar, and Y. Zhu, “RoboCasa: Large-scale simulation of everyday tasks for generalist robots,” in *Robotics: Science and Systems (RSS)*, 2024.
- [27] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: A physics engine for model-based control,” in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Vilamoura-Algarve, Portugal: IEEE, Oct. 2012, pp. 5026–5033.